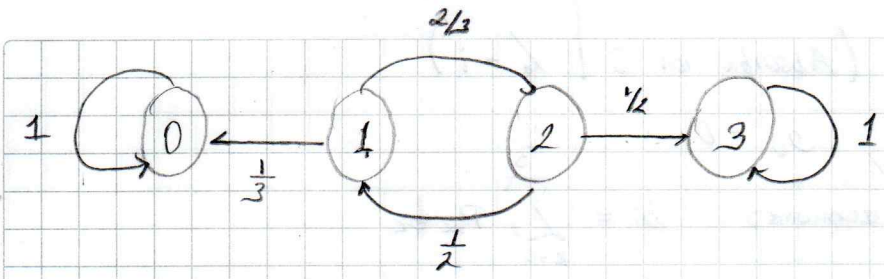
Clase.

20 / 02 / 26



$S = \{0,1,2,3\} \xrightarrow{S_k}$ Classes: $S_1 = \{0\} \cup S_2 = \{1,2\} \cup S_3 = \{3\}$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/3 & 0 & 2/3 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Probabilidades de absorcion.
- $a_i = \mathbb{P}(\text{Absorba en } 0 \mid X_0 = i)$
- $a_0 = 1$, $a_3 = 0$
- $a_1 = a_0 P_{10} + a_1 P_{11} + a_2 P_{12} + a_3 P_{13}$

- Regla de la Probabilidad total: $\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\, \mathbb{P}(B_i)$

↳ Evento Condicionado: $A_i = \{\text{Absorba en } 0 \mid X_0 = i\}$

Para $B_i = \{X_1 = k \mid X_0 = i\}$

↳ formula Recursiva: $a_i = \sum_k a_k P_{ik}$.

- observacion

$a_0 P_{10}$ de ir de 1 a cero

$0 \xleftarrow{1/2} 1$

- $a_1 = \frac{1}{3} a_0 + 0 a_1 + \frac{2}{3} a_2 + 0 a_3$
- $a_2 = a_0 P_{20} + a_1 P_{21} + a_2 P_{22} + a_3 P_{23}$

↳ $a_2 = 0 a_0 + \frac{1}{2} a_1 + 0 a_2 + \frac{1}{2} a_3$

the values of a_1 and a_2. Using Equation:

$a_0 = a_0 = 1$

$a_1 = \frac{1}{3} a_0 + \frac{2}{3} a_2$

$a_2 = \frac{1}{2} a_1 + \frac{1}{2} a_3$

$a_3 = a_3 = 0$

$a_1 = 1/2$, $a_2 = 1/4$.

"Sanity Check"

- Ejercicio: $b_i = \mathbb{P}(\text{Absorba en } 3 \mid X_0 = i)$

$$b_3 = 1 \quad , \quad b_0 = 0$$

- Planteamiento de las ecuaciones $b_i = \sum_{k=0}^{3} P_{ik} b_k$

- $b_1 = P_{10} b_0 + P_{11} b_1 + P_{12} b_2 + P_{13} b_3$

$\hookrightarrow b_1 = \left(\frac{1}{3}\right)(0) + (0) b_1 + \left(\frac{2}{3}\right) b_2 + (0)(1) \Rightarrow b_1 = \frac{2}{3} b_2$

- $b_2 = P_{20} b_0 + P_{21} b_1 + P_{22} b_2 + P_{23} b_3$

$\hookrightarrow b_2 = (0)(0) + \left(\frac{1}{2}\right) b_1 + (0) b_2 + \frac{1}{2}(1) \Rightarrow b_2 = \frac{1}{2} b_1 + \frac{1}{2}$

- Resolver el sistema de ecuaciones

$b_1 = \frac{2}{3} b_2$ y $b_2 = \frac{1}{2} b_1 + \frac{1}{2} \Rightarrow b_2 = \frac{1}{2}\left(\frac{2}{3} b_2\right) + \frac{1}{2}$

$b_2 = \frac{2}{6} b_2 + \frac{1}{2}$, $b_2 = \frac{1}{3} b_2 + \frac{1}{2}$ despejamos b_2

$b_2 - \frac{1}{3} b_2 = \frac{1}{2} \Rightarrow \frac{2}{3} b_2 = \frac{1}{2} \Rightarrow b_2 = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$

$\rightarrow b_1 = \frac{2}{3}\left(\frac{3}{4}\right) = \frac{6}{12} = \frac{1}{2}$

- Conclusión y Comprobación ("Sanity Check")

$b_0 = 0$

$b_1 = 1/2$

$b_2 = 3/4$

$b_3 = 1$